Esercizi

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1. *Semplifica le seguenti espressioni utilizzando i prodotti notevoli:*

   (a) $(a-1)^2(a+2)+(1-a)(a+1)(a-2)$

   (b) $[(x-1)(x+1)-(1-2x)(1+2x)]^2-(5x^2+1)^2+(-6x)(-5x)$

   (c) $(x^2+1)^3-(x^3+1)^2+3x^2(x+1)(x-1)-(-3x^2)(-2x^2)$

   (d) $(x^3+x^2+x)-(x^2+x^3)^2-(x^2+x)^2$

   (e) $(a^3+a^2+1)^2-(a^2-1)^3-4(a^2+1)^2+(3a+1)(3a-1)+3$

   (f) $[(x+1)^2-2x]^2-(x+1)^2(x-1)^2$

   (g) $(a^2-a-1)^2-(a+1)^2(a-1)^2-a^2(1-2a)$

   (h) $(a+2)^2(a-2)^2-(a^3-a-2)^2-a(2a-2)^2+a(a+8-3a^3)+(a^3-4)(a^3+4)$

2. *Risolvi le seguenti equazioni*

   (a) $(3x-2)^2-(3x-4)(3x+4)=(2x+1)^2-(-2x)^2$

   (b) $(2-x)(2+x)(4-x^2)=x^2(x-2)(x+2)-(2x-1)(2x+3)$

   (c) $(x+2)^3-(x+1)^3=(2x-1)(2x+1)-x^2$

   (d) $(x^2-2x+3)^2=(x^2-2)(x^2+2)+2x^2(5-2x)$

   (e) $1-\frac{1}{2}\left[3-\frac{1}{3}(x-4)\right]=\frac{1}{2}(x-2)-\frac{1}{3}(3-x)$

   (f) $\frac{x(x-1)}{2}-\frac{(x-1)^2}{5}+\frac{(x-2)(x+3)}{10}=\frac{2}{5}x^2-x-\frac{9}{5}$

   (g) $\left(\frac{x-3}{2}\right)^2-\left(\frac{x-2}{3}\right)^2=5\left(\frac{1}{6}x-1\right)\left(\frac{1}{6}x+1\right)-\frac{1}{18}x-\frac{7}{36}$